Optimizing S-curve Velocity for Motion Control

Xiaolin Li
Department of Mathematics and Computer Science
Alabama State University, Montgomery, AL 36104, USA
xli@alasu.edu

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Abstract

Velocity profile design plays an important role in motion control technology. Among various velocity profiles, the S-curve velocity profile is a smooth and continuous curve and is differentiable to the second order. In this article, we discuss how to optimize the S-curve velocity given the constraints on jerk, acceleration and velocity magnitude regarding the physical capacity of the motion devices of each axis. The optimized S-curve velocity profile can be used to program motion processors to produce a smooth, fast and accurate motion.

Index terms: point-to-point motion, physical constraints, S-curve velocity

1 Introduction

Velocity profile design plays an important role in motion control technology. In a conventional close-chain system, a motion is controlled by a controller and its velocity profile is implicitly defined by the system. The controller takes feedback from sensors in different chains regarding movement position, speed, accelaration, jerk, and so on. Within each chain, the actual state is compared with the desired state and then the controller adjusts the system output to eliminate various errors. Meantime, the controller optimizes the system response time and ensures the system stability. Based on various feedback, the system produces an optimal velocity profile corresponding to a smooth, fast and accurate motion [1, 3, 4, 6, 7, 8, 9, 10].

In an open-chain system, the velocity profile of a motion is explicitly designed and calculated [2, 5, 10]. Since there is no feedback involved in the control, the motion is completely defined by its velocity. In a coordinated motion, one must consider the physical constraints on jerk, acceleration and velocity magnitude of each axis because axis devices such as servomotors have their capacity limitation. Given a start position and an end position of a motion in a 3D space, ideally, we want to use the axis devices to their capacity extreme so that the motion will be smooth, fast and accurate. Among various options, the S-curve velocity is one of the best choices. An S-curve velocity profile is a smooth and continuous curve and is differentiable to the second order. We can easily incorporate all physical constraints regarding jerk, acceleration and velocity magnitude into consideration.

In this article, we discuss the fundamentals of point-to-point motion using S-curve velocity. Given the physical constraints on jerk, acceleration, and velocity magnitude of each axis, we show how to calculate the different edges in an S-curve velocity profile and how to optimize the S-curve velocity profile. The optimized S-curve velocity profile can then be used to program motion processors to get a smooth, fast and accurate motion.

The remainder of the paper is organized as follows: Section 2 discusses 3D velocity decomposition and 1D constraint synthesis. Section 3 presents the calculation of an S-curve velocity under given conditions. Section 4 describes a binary search method for the S-curve velocity optimization. Finally, Section 5 concludes this paper.

2 Velocity Decomposition and Constraint Synthesis

2.1 Velocity decomposition

Let us consider a straight movement given a start point $p_s(x_s, y_s, z_s)$ and an end point $p_e(x_e, y_e, z_e)$ in a 3D space. The directional numbers and the directional angles of the velocity are given by

$$\begin{aligned} \Delta x &= x_e - x_s \\ \Delta y &= y_e - y_s \\ \Delta z &= z_e - z_s \end{aligned} \tag{1}$$

$$\begin{aligned} \cos\alpha &= \frac{\Delta x}{l} \\ \cos\beta &= \frac{\Delta y}{l} \\ \cos\gamma &= \frac{\Delta z}{l} \end{aligned} \tag{2}$$

where l is the distance between $p_s(x_s, y_s, z_s)$ and $p_e(x_e, y_e, z_e)$ such that

$$l = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} \tag{3}$$

Let $v(t)$ be the velocity magnitude of the movement, which is a function of time. Suppose the movement starts from $p_s(x_s, y_s, z_s)$ at time $t = 0$ with $v(0) = v_s$, and reaches $p_e(x_e, y_e, z_e)$ at time $t = T$ with $v(T) = v_e$, then $v(t)$ must satisfy the travel distance constraint such that

$$\int_0^T v(t)dt = l \tag{4}$$

Since the movement takes place in the 3D space, $v(t)$ can be decomposed into 1D component velocity magnitudes $x(t)$, $y(t)$ and $z(t)$ such that

$$\begin{aligned} x(t) &= v(t)\cos\alpha \\ y(t) &= v(t)\cos\beta \\ z(t) &= v(t)\cos\gamma \end{aligned} \tag{5}$$

where it satisfies

$$v(t) = \sqrt{x^2(t) + y^2(t) + z^2(t)} \tag{6}$$

And the above component velocity magnitudes can be used to program axis processors.

2.2 Constraint Synthesis

Let us consider the physical constraints on each axis. Because a motion system may have different physical structures, we may have different constraints on different axes. Let x_{max}, y_{max} and z_{max} be the upper bounds of the velocity magnitude, x'_{max}, y'_{max} and z'_{max} be the upper bounds of the acceleration, and $x"_{max}$, $y"_{max}$ and $z"_{max}$ be the upper bounds of the jerk of individual axes such that

$$
\begin{aligned}
x(t) &\leq x_{max} \\
y(t) &\leq y_{max} \\
z(t) &\leq z_{max}
\end{aligned}
\tag{7}
$$

$$
\begin{aligned}
x'(t) &\leq x'_{max} \\
y'(t) &\leq y'_{max} \\
z'(t) &\leq z'_{max}
\end{aligned}
\tag{8}
$$

$$
\begin{aligned}
x"(t) &\leq x"_{max} \\
y"(t) &\leq y"_{max} \\
z"(t) &\leq z"_{max}
\end{aligned}
\tag{9}
$$

where $'$ and $"$ denote the first order derivative (corresponding to acceleration) and the second order derivative (corresponding to jerk) of a function, respectively. Then the synthesis of the above 1D constraints on the 3D velocity is given by

$$
\begin{aligned}
&v(t) \leq v_{max} \\
&|v'(t)| \leq v'_{max} \\
&|v"(t)| \leq v"_{max}
\end{aligned}
\tag{10}
$$

where

$$
\begin{aligned}
v_{max} &= \min\{\frac{x_{max}}{|\cos\alpha|}, \frac{y_{max}}{|\cos\beta|}, \frac{z_{max}}{|\cos\gamma|}\} \\
v'_{max} &= \min\{\frac{x'_{max}}{|\cos\alpha|}, \frac{y'_{max}}{|\cos\beta|}, \frac{z'_{max}}{|\cos\gamma|}\} \\
v"_{max} &= \min\{\frac{x"_{max}}{|\cos\alpha|}, \frac{y"_{max}}{|\cos\beta|}, \frac{z"_{max}}{|\cos\gamma|}\}
\end{aligned}
\tag{11}
$$

Bearing the above constraints in mind, we can design a velocity profile $v(t)$ for the straight movement from $p_s(x_s, y_s, z_s)$ to $p_e(x_e, y_e, z_e)$ in time interval $[0, T]$. Except for the above physical constraints and the travel distance constraint, $v(t)$ also satisfies the following boundary conditions:

$$
\begin{aligned}
0 \leq v(0) = v_s < v_{max} \\
0 \leq v(T) = v_e < v_{max}
\end{aligned}
\tag{12}
$$

We want the movement to be smooth, fast and accurate. Among various options, the S-curve velocity is one of the best choices to achieve our goals.

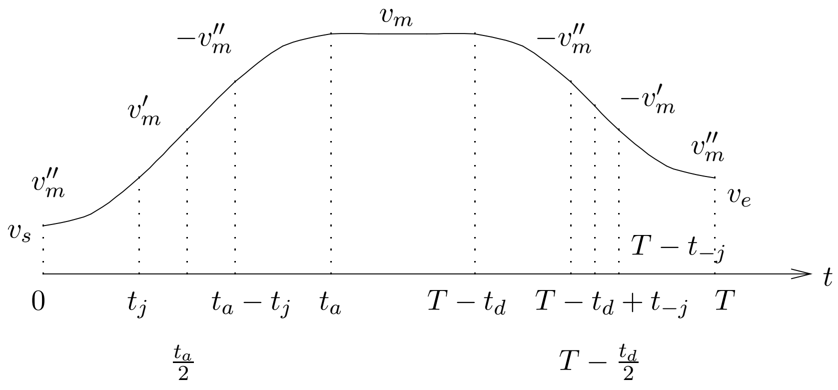

Figure 1: S-curve velocity profile

3 S-curve Velocity Profile

An S-curve velocity profile is differentiable to second order, where both the velocity magnitude and the acceleration are continuous functions. Let $v(t)$ be an S-curve velocity profile defined in time interval $[0, T]$, then $\forall t \in [0, T]$, we have

$$\begin{aligned} v'(t) &= v'(0) + \int_0^t v"(w)dw \\ v(t) &= v(0) + \int_0^t v'(w)dw \end{aligned} \tag{13}$$

$v(t)$ also satisfies some boundary conditions. Let $v(0) = v_s$ and $v(T) = v_e$, for calculation convenience, $v(t)$ can be decomposed into three edges: a) an ascending edge in time interval $[0, t_a]$, where the velocity magnitude increases from v_s to v_m, and t_a is the ascending time of the velocity magnitude, b) a descending edge in time interval $[T - t_d, T]$, where the velocity magnitude decreases from v_m to v_e, and t_d is the descending time of the velocity magnitude, and c) a horizontal edge in time interval $(t_a, T - t_d)$, where the velocity magnitude remains a constant value v_m, See Figure 1.

Given a start point and an end point with physical constraints and boundary conditions as described, we choose $v"_m = v"_{max}$ in our solution to an S-curve velocity profile so that whenever applicable, the acceleration or deceleration can change quickly. Then we try to find an optimal value v_m under the travel distance constraint and the conditions $v_s < v_m \leq v_{max}$ and $v_e < v_m \leq v_{max}$, which gives the solutions to the ascending edge, the descending edge, and the horizontal edge.

3.1 Ascending edge

The ascending edge of an S-curve velocity is symmetrical about its center $(\frac{t_a}{2}, \frac{v_m + v_s}{2})$. Whenever applicable, the time interval $[0, t_a]$ can be divided into three regions corresponding to three phases: 1) positive jerk region $[0, t_j)$, 2) constant acceleration region $[t_j, t_a - t_j]$, and 3) negative jerk region

$(t_a - t_j, t_a]$, such that

$$\begin{aligned} & v(0) = v_s \\ & v(t_a) = v_m \end{aligned} \tag{14}$$

$$\begin{aligned} & v'(0) = 0 \\ & v'(t_a) = 0 \\ & v'(t) = v'_m, \quad t_j \le t \le t_a - t_j \end{aligned} \tag{15}$$

$$v"(t) = \begin{cases} v"_m, & 0 < t < t_j \\ 0, & t_j \le t \le t_a - t_j \\ -v"_m, & t_a - t_j < t < t_a \end{cases} \tag{16}$$

where v_m, v'_m and $v"_m$ are constants, satisfying $v_s < v_m \le v_{max}$, $v'_m \le v'_{max}$, and $v"_m = v"_{max}$. We can solve the ascending edge in the following steps:

Firstly, consider the case where $t_j = \dfrac{t_a}{2}$, i.e.

$$v"(t) = \begin{cases} v"_{max}, & 0 < t < \frac{t_a}{2} \\ -v"_{max}, & \frac{t_a}{2} < t < t_a \end{cases} \tag{17}$$

Notice that $v(t)$ and $v'(t)$ are continuous functions and that $v(0) = v_s$ and $v'(0) = 0$, when $0 \le t \le \dfrac{t_a}{2}$, we have

$$\begin{aligned} v'(t) &= \int_0^t v"_{max} dw \\ &= v"_{max} t \end{aligned} \tag{18}$$

$$\begin{aligned} v(t) &= v(0) + \int_0^t v'(w) dw \\ &= v_s + \frac{1}{2} v"_{max} t^2 \end{aligned} \tag{19}$$

$$\begin{aligned} v(\frac{t_a}{2}) &= v_s + \frac{1}{8} v"_{max} t_a^2 \\ &= \frac{v_m + v_s}{2} \end{aligned} \tag{20}$$

It follows that

$$t_a = 2\sqrt{\frac{v_m - v_s}{v"_{max}}} \tag{21}$$

Secondly, consider the maximum acceleration achieved at $t = t_j = \dfrac{t_a}{2}$. Notice that

$$\begin{aligned} v'(\frac{t_a}{2}) &= \frac{1}{2} v"_{max} t_a \\ &= \sqrt{v"_{max}(v_m - v_s)} \end{aligned} \tag{22}$$

we have two possible cases:

1. In the case that

$$\sqrt{v"_{max}(v_m - v_s)} \leq v'_{max} \tag{23}$$

the acceleration constraint has been satisfied.

2. In the case that

$$\sqrt{v"_{max}(v_m - v_s)} > v'_{max} \tag{24}$$

the jerk regions must be adjusted such that $t_j < \dfrac{t_a}{2}$. To resolve t_j and t_a, consider t_j with the condition that $v'(t_j) = v'_{max}$. From equations (18) and (19), we have

$$\begin{aligned} v'(t_j) &= v"_{max} t_j \\ &= v'_{max} \end{aligned} \tag{25}$$

$$t_j = \frac{v'_{max}}{v"_{max}} \tag{26}$$

$$\begin{aligned} v(t_j) &= v_s + \frac{1}{2} v"_{max} t_j^2 \\ &= v_s + \frac{v'^2_{max}}{2v"_{max}} \end{aligned} \tag{27}$$

and it follows that

$$v(t_a - t_j) = v_m - \frac{v'^2_{max}}{2v"_{max}} \tag{28}$$

since the ascending edge is symmetrical about $(\dfrac{t_a}{2}, \dfrac{v_m + v_s}{2})$.

On the other hand, when $t_j \leq t \leq t_a - t_j$, we have

$$v(t) = v'_{max}(t - t_j) + v_s + \frac{v'^2_{max}}{2v"_{max}} \tag{29}$$

$$\begin{aligned} v(t_a - t_j) &= v'_{max}(t_a - 2t_j) + v_s + \frac{v'^2_{max}}{2v"_{max}} \\ &= v'_{max} t_a + v_s - \frac{3v'^2_{max}}{2v"_{max}} \end{aligned} \tag{30}$$

In the light of equations (28) and (30), we have

$$t_a = \frac{v_m - v_s}{v'_{max}} + \frac{v'_{max}}{v''_{max}} \tag{31}$$

Thirdly, let us consider the travel distance in the time interval $[0, t_a]$ after t_j and t_a have been solved. In either case, we have

$$\int_0^{t_a} v(t)dt = \frac{1}{2}(v_m + v_s)t_a \tag{32}$$

3.2 Descending edge

Similar to the ascending edge, the descending edge of an S-curve velocity is symmetrical about its center $(T - \frac{t_d}{2}, \frac{v_m + v_e}{2})$. Whenever applicable, the time interval $[T - t_d, T]$ can be divided into three regions corresponding to three phases: 1) negative jerk region $[T - t_d, T - t_d + t_{-j})$, 2) constant deceleration region $[T - t_d + t_{-j}, T - t_{-j}]$, and 3) positive jerk region $(T - t_{-j}, T]$, such that

$$\begin{array}{l} v(T - t_d) = v_m \\ v(T) = v_e \end{array} \tag{33}$$

$$\begin{array}{ll} v'(T - t_d) = 0 & \\ v'(T) = 0 & \\ v'(t) = -v'_m, & T - t_d + t_{-j} \le t \le T - t_{-j} \end{array} \tag{34}$$

$$v''(t) = \begin{cases} -v''_m, & T - t_d < t < T - t_d + t_{-j} \\ 0, & T - t_d + t_{-j} \le t \le T - t_{-j} \\ v''_m, & T - t_{-j} < t < T \end{cases} \tag{35}$$

Given the conditions that $v_e < v_m \le v_{max}$, $v'_m \le v'_{max}$ and $v''_m = v''_{max}$, we can solve the descending edge in a way similar to solving the ascending edge.

Firstly, consider the case where $t_{-j} = \frac{t_d}{2}$, i.e.

$$v''(t) = \begin{cases} -v''_{max}, & T - t_d < t < T - \frac{t_d}{2} \\ v''_{max}, & T - \frac{t_d}{2} < t < T \end{cases} \tag{36}$$

When $T - t_d \le t \le T - \frac{t_d}{2}$, we have

$$\begin{aligned} v'(t) &= -\int_{T-t_d}^{t} v''_{max} dw \\ &= -v''_{max}(t - T + t_d) \end{aligned} \tag{37}$$

$$\begin{aligned} v(t) &= v_m - \int_{T-t_d}^{t} v''_{max}(w - T + t_d) dw \\ &= v_m - \frac{1}{2} v''_{max}[t^2 - (T - t_d)^2] + v''_{max}(T - t_d)[t - (T - t_d)] \end{aligned} \tag{38}$$

$$\begin{aligned} v(T - \frac{t_d}{2}) &= v_m - \frac{1}{8} v''_{max} t_d^2 \\ &= \frac{v_m + v_e}{2} \end{aligned} \tag{39}$$

$$t_d = 2\sqrt{\frac{v_m - v_e}{v''_{max}}} \tag{40}$$

Secondly, consider the maximum deceleration achieved at $t = T - t_d + t_{-j} = T - \frac{t_d}{2}$. Because

$$\begin{aligned} v'(T - \frac{t_d}{2}) &= -\frac{1}{2} v"_{max} t_d \\ &= -\sqrt{v"_{max}(v_m - v_e)} \end{aligned} \tag{41}$$

we have two possible cases:

1. In the case that

$$\sqrt{v"_{max}(v_m - v_e)} \leq v'_{max} \tag{42}$$

the deceleration constraint has been satisfied.

2. In the case that

$$\sqrt{v"_{max}(v_m - v_e)} > v'_{max} \tag{43}$$

the jerk regions must be adjusted such that $t_{-j} < \frac{t_d}{2}$. To resolve t_{-j} and t_d, consider t_{-j} with the condition that $v'(T - t_d + t_{-j}) = -v'_{max}$. From equations (37) and (38), we have

$$\begin{aligned} v'(T - t_d + t_{-j}) &= -v"_{max} t_{-j} \\ &= -v'_{max} \end{aligned} \tag{44}$$

$$t_{-j} = \frac{v'_{max}}{v"_{max}} \tag{45}$$

$$\begin{aligned} v(T - t_d + t_{-j}) &= v_m - \frac{1}{2} v"_{max} t_{-j}^2 \\ &= v_m - \frac{v'^2_{max}}{2v"_{max}} \end{aligned} \tag{46}$$

and it follows that

$$v(T - t_{-j}) = v_e + \frac{v'^2_{max}}{2v"_{max}} \tag{47}$$

since the descending edge is symmetrical about $(T - \frac{t_d}{2}, \frac{v_m + v_e}{2})$.

On the other hand, when $T - t_d + t_{-j} \leq t \leq T - t_{-j}$, we have

$$v(t) = v_m - \frac{v'^2_{max}}{2v"_{max}} - v'_{max}[t - (T - t_d + t_{-j})] \tag{48}$$

$$\begin{aligned} v(T - t_{-j}) &= v_m - \frac{v'^2_{max}}{2v"_{max}} - v'_{max}(t_d - 2t_{-j}) \\ &= v_m + \frac{3v'^2_{max}}{2v"_{max}} - v'_{max} t_d \end{aligned} \tag{49}$$

In the light of the equations (47) and (49), we have

$$t_d = \frac{v_m - v_e}{v'_{max}} + \frac{v'_{max}}{v"_{max}} \tag{50}$$

Thirdly, consider the travel distance in the time interval $[T - t_d, T]$ once t_{-j} and t_d have been solved. In either case, we have

$$\int_{T-t_d}^{T} v(t)dt = \frac{1}{2}(v_m + v_e)t_d \tag{51}$$

3.3 Horizontal edge and travel distance constraint

Let us consider how to choose v_m given the condition that $v_m \leq v_{max}$ and the travel distance constraint that

$$\begin{aligned} \int_0^T v(t)dt &= \int_0^{t_a} v(t)dt + \int_{t_a}^{T-t_d} v(t)dt + \int_{T-t_d}^{T} v(t)dt \\ &= l \end{aligned} \tag{52}$$

Suppose we have an initial value v_m with related solutions to the ascending edge and the descending edge, then there are two possible cases regarding the travel distance constraint:

1. In the case that

$$\int_0^{t_a} v(t)dt + \int_{T-t_d}^{T} v(t)dt \leq l \tag{53}$$

we have

$$v_m(T - t_a - t_d) = l - \frac{1}{2}[(v_m + v_s)t_a + (v_m + v_e)t_d] \tag{54}$$

$$T = t_a + t_d + \frac{1}{v_m}\{l - \frac{1}{2}[(v_m + v_s)t_a + (v_m + v_e)t_d]\} \tag{55}$$

And the S-curve velocity profile is solved with the travel distance constraint satisfied.

2. In the case that

$$\int_0^{t_a} v(t)dt + \int_{T-t_d}^{T} v(t)dt > l \tag{56}$$

we must reduce the value v_m and re-calculate the ascending edge and the descending edge so that the travel distance constraint can hold. We can solve and optimize an S-curve velocity profile using the following binary search method.

4 Binary search method

In summary of our previous discussion, given the maximum velocity magnitude v_{max}, the maximum acceleration v'_{max}, and the maximum jerk $v"_{max}$, with the boundary conditions that $v(0) = v_s$ and $v(T) = v_e$, t_j, t_a, t_{-j} and t_d are functions of v_m and may have different expressions:

$$t_j = \begin{cases} \sqrt{\dfrac{v_m - v_s}{v"_{max}}}, & \sqrt{v"_{max}(v_m - v_s)} \leq v'_{max} \\ \dfrac{v'_{max}}{v"_{max}}, & \sqrt{v"_{max}(v_m - v_s)} > v'_{max} \end{cases} \tag{57}$$

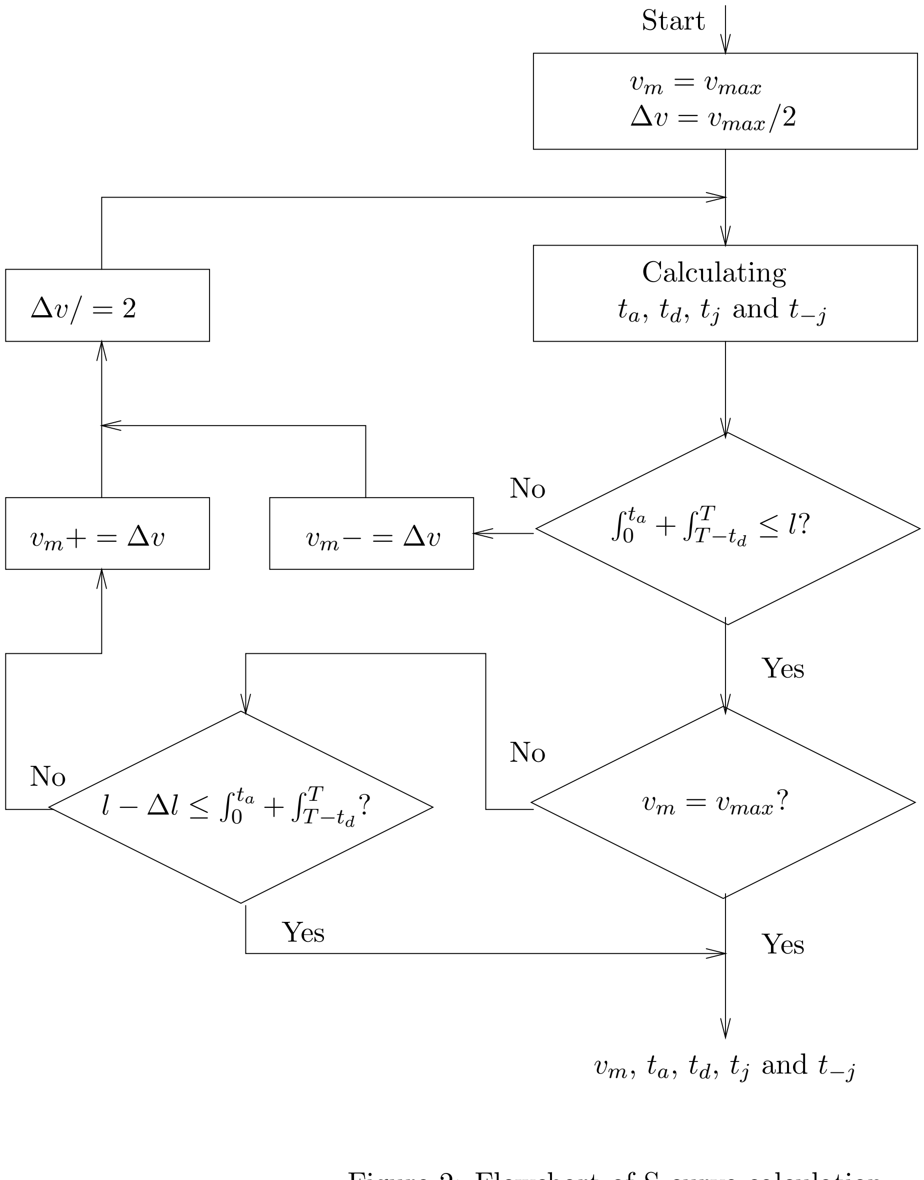

Figure 2: Flowchart of S-curve calculation

$$
t_a = \begin{cases} 2t_j, & \sqrt{v"_{max}(v_m - v_s)} \le v'_{max} \\ \dfrac{v_m - v_s}{v'_{max}} + t_j, & \sqrt{v"_{max}(v_m - v_s)} > v'_{max} \end{cases} \tag{58}
$$

$$
t_{-j} = \begin{cases} \sqrt{\dfrac{v_m - v_e}{v"_{max}}}, & \sqrt{v"_{max}(v_m - v_e)} \le v'_{max} \\ \dfrac{v'_{max}}{v"_{max}}, & \sqrt{v"_{max}(v_m - v_e)} > v'_{max} \end{cases} \tag{59}
$$

$$
t_d = \begin{cases} 2t_{-j}, & \sqrt{v"_{max}(v_m - v_e)} \le v'_{max} \\ \dfrac{v_m - v_e}{v'_{max}} + t_{-j}, & \sqrt{v"_{max}(v_m - v_e)} > v'_{max} \end{cases} \tag{60}
$$

Bearing the above equations in mind, we can solve an S-curve velocity profile under given conditions and optimize the S-curve velocity profile by the binary search method illustrated in Figure 2. Based on the fact that a large velocity magnitude ensures a fast motion, we choose $v_m = v_{max}$ and get initial solutions to the ascending edge and the descending edge in the first place. Then we test the travel distance constraint and modify v_m if necessary. In Figure 2, Δl is a threshold for termination judgment. When the algorithm terminates, v_m is fixed and the solution to an optimal S-curve velocity profile readily follows.

5 Conclusion

In this article we have presented a technique of point-to-point motion using S-curve velocity. Given the start point and the end point of the streight movement in a 3D space with constraints on jerk, acceleration and velocity magnitude of each component axis, we have shown how to decompose the 3D velocity into 1D component velocities and how to synthesize 1D axis constraints on the 3D movement. We have chosen to use the S-curve velocity profile because it is a smooth and continuous curve and is differentiable to the second order. We have shown how to calculate the ascending edge, the horizontal edge, and the descending edge in an S-curve velocity profile given the constraits on the jerk, the accelaration and the velocity magnitude of each axis. We have also presented a binary search method to optimize the S-curve velocity profile so that the motion will be smooth, fast and accurate.

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